

ALEX AGUILAR VIEL

Aerospace Engineering · Aerodynamics & Motorsport

CONTACT

alexaguilar.aa245@gmail.com

Alex Aguilar Viel

+34 673 704 247

Valencia, Spain

SKILLS

CFD & Aerodynamic Simulation	2+ years
CAD Design	2+ years
Python & MATLAB	2+ years
C++	1 year
Oracle SQL Developer	1+ years
Power BI & Dashboards	1+ years
Power Automate & AI	1 year
MRO Systems	1+ years

LANGUAGES

Spanish	Native
Valencian	Native
English	B2

ABOUT ME

Aerospace Engineering graduate (Aerospace Vehicles specialisation) passionate about competition aerodynamics, CFD simulation, and data-driven engineering. Currently working as a Systems Engineer at Air Nostrum, contributing to the digitalization of aircraft maintenance operations through process automation, AI-based tools, and engineering analytics.

Core Expertise:

- **Aerodynamics & CFD:** Multi-element surface design, Star CCM+ simulation, and MATLAB post-processing. Developed a numerical methodology for F1 front & rear wing optimisation under the 2026 regulation in my Final Degree Project (9.2/10).
- **Data & Automation:** Oracle SQL, Power BI engineering dashboards, and Power Automate workflows applied in real operational airline environments.
- **Engineering & Systems:** Bridging aerospace fundamentals with digital solutions – from CAD design to MRO process improvement in a live operational setting.

Driven by precision, continuous improvement, and the ambition to bring engineering rigour into high-performance motorsport.

WORK EXPERIENCE

Junior Systems & IT Engineer Apr 25 – Sept 26

Air Nostrum Airlines

MRO digitalization and data-driven process improvement, combined with full-time studies (30 h/week).

- **eMobility:** Moved maintenance tasks to tablet-based, paperless workflows.
- **Automation:** Led Power Automate and AI-based workflow automation.
- **Analytics:** Built Power BI dashboards for maintenance KPIs.
- **Data:** Queried and analysed MRO data with Oracle SQL.

WORK EXPERIENCE (CONT.)

Aerodynamics Member Sept 23 – Feb 24
FSUPV Formula Student Team – UPV

Design cycle of aerodynamic surfaces for a Formula Student prototype.

- **CFD:** Star-CCM+ simulations (RANS k-ω SST) with Java macros to automate iterations.
- **CAD:** Aerodynamic surfaces in Siemens NX under FS regulations.
- **Validation:** Correlated CFD with track data at Circuito Ricardo Tormo.
- **Post-processing:** MATLAB scripts for performance benchmarking.

EDUCATION

MSc in Advanced Motorsport Engineering Oct 26 – Present
Universidad Politécnica de Madrid (UPM)

Postgraduate programme in race car engineering: aerodynamics, vehicle dynamics and powertrain.

BSc in Aerospace Engineering – Aerospace Vehicles Specialisation Sept 22 – Sept 26
Universitat Politècnica de València (UPV)

Graduated September 2026. Average grade: 7.0/10.

- **Final Degree Project (TFG) – Grade 9.2/10:** *"Development of a numerical methodology to design and optimise a multi-element lifting surface in different applications"* – CFD design and optimisation of a 2026-regulation F1 rear wing, studying the aerodynamic impact of replacing DRS with the new active Z and X deployment modes in both cornering and straight-line conditions (Star-CCM+ & CATIA V5).

Pre-selection – Valencia Physics Olympiad (2022) Nov 21 – Feb 22
Universitat de València

Selected as 1 of 22 students out of 500+ candidates in the Valencia regional phase. Completed an intensive 72-hour university-level preparation course covering classical mechanics, electromagnetism, thermodynamics, optics, and modern physics, followed by individual written exams under competition conditions.

Scientific-Technical Baccalaureate Sept 20 – Jun 22
IES La Encarnación – Sueca, Valencia

- **Academic Distinction:** Graduated with an overall average grade of 9.89/10.
- **University Entrance:** Achieved a University entrance score of 13.452/14 in the PAU exams.

TECHNICAL SKILLS

Programming & Data
Python | MATLAB | C++
Oracle SQL Developer
Power BI | AI Tools

CFD & Simulation
Star-CCM+
Aerodynamic analysis
Post-processing | Java Macros

CAD & Engineering
Siemens NX | CATIA V5
SolidWorks | Autodesk Fusion

CERTIFICATES

- **First Certificate in English B2.** Cambridge
- **Official Valencian Language Proficiency Certificate, level C1:** Generalitat Valenciana
- **Distinction with Honors in Secondary Education.** IES La Encarnación, Valencia
- **First Aid & AED:** Global Encesap